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A Power Optimized 32-Bit Carry Look Ahead Adder Employing PTL and NMOS Power Gated Technique for VLSI Circuits

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***Abstract*—This paper presents the design and implementation of a 32-bit Carry Look-Ahead (CLA) Adder using Pass Transistor Logic (PTL) combined with a Power Gated NMOS technique to achieve enhanced power efficiency in deep sub-micron technology. The proposed architecture is constructed using modular 4-bit CLA blocks and hierarchically expanded to form a 32-bit structure. To reduce leakage power during idle conditions, a footer Power Gated NMOS transistor is incorporated between the logic circuit and ground, effectively cutting off the leakage current path in standby mode. The PTL based implementation reduces transistor count and switching capacitance, thereby minimizing dynamic power consumption, while the Power Gated NMOS mechanism significantly suppresses static power dissipation. The entire design is implemented using Tanner EDA tool in 22nm technology. The proposed design reduces total power consumption to 23.01% compared to conventional PTL design. Although a slight increase in transistor count is observed due to the inclusion of the Power Gated NMOS structure, the overall energy efficiency and power-delay performance are significantly improved. Therefore, the proposed 32-bit PTL based CLA with Power Gated NMOS technique is highly suitable for VLSI circuits.**

Keywords—Carry Look-Ahead Adder (CLA), Pass Transistor Logic (PTL), NMOS Power Gating Transistor, Leakage Power Reduction, Low-Power VLSI Design

I. INTRODUCTION

Adders constitute one of the most fundamental and performance-critical building blocks in digital integrated circuits, forming the core of Arithmetic Logic Units (ALUs), digital signal processors (DSPs), microprocessors, and high-performance computing systems. The overall computational speed and energy efficiency of these systems are strongly influenced by the performance of the underlying adder architecture [1]. Among various adder techniques, the Carry Look-Ahead (CLA) adder is widely recognized for its high-speed capability, as it mitigates the carry propagation delay inherent in Ripple Carry Adders (RCAs). Unlike RCAs, which compute carry signals sequentially, the CLA architecture generates propagate and generate signals in parallel, thereby significantly reducing critical path delay and improving overall computational throughput.

Pass Transistor Logic (PTL) has received significant interest as an effective alternative to traditional static CMOS logic because it has a smaller number of transistors, less input capacitance, and less switching activity. PTL based circuits have small implementations and high-performance in speed by removing redundant pull-up and pull-down networks [3]. As of late, it has been seen that PTL based arithmetic circuits have the potential to provide improved power-delay product (PDP) attributes and retain a competitive operational speed [3], [4]. PTL design can however be susceptible to loss in threshold voltage, and decrease in voltage swing especially in aggressively scaled technologies which could adversely impact signal integrity as well as raise leakage currents. Hence, we need complementary techniques to leakage mitigation to completely enjoy the benefits of PTL logic.

The conventional 32-bit CLA using PTL with transmission gate suffer with large power consumption and delay. To overcome those limitations, this paper proposed a 32-bit CLA using Pass Transistor Logic with power gated NMOS technique realize a high level of reduction in total power consumption, enhanced power-delay product, as well as improved energy efficiency by combining the PTL powered dynamic power reduction and the NMOS powered leakage suppression. The proposed 32-bit adder is hierarchical with modular CLA block of 4-bits to guarantee scalability, regularity of design and implementation ease. Therefore, the proposed architecture can be used in several low-power, high-performance VLSI applications like ALU's, DSP's, Embedded and IOT Devices at deep sub-micron technology nodes.

The remaining sections of this paper are structured in the following way. Section I outlines the introduction to various adders, Section II outlines the already Conventional others made carrying research work to the concept of carry look ahead adder. Section III talked about the existing methodology approach concerning PTL Based 32-Bit CLA, section IV focus on the proposed methodology concerning the implementation of 32-Bit CLA using PTL and Power Gated NMOS technique. In section V, results of simulation and discussions were explained depending on area parameters, delay and power consumption. And lastly section VI gives conclusion concerning proposed system with regards to existing system.

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II. LITERATURE REVIEW

M and R. M. Bommi, author et al. proposed 4-bit CLA using PTL based Transmission gate approach to reduce the number of transistor to 58.82% and power consumption to 28.14% by using cadence virtuoso tool compared with conventional dynamic CMOS and Static CMOS [11]. S. M. S. Samsudeen, I. Jaiswal, author et al. proposed 4-bit carry generator circuit using PTL based 2X1 multiplexer method to reduce the delay, power consumption and improve the circuit efficiency [12]. Eresh, N. Yadav author et al. proposed 32-bit carry look ahead adder using pipeline approach to improve the throughput and reduce the critical path delay [13]. J. Barman and V. Kumar author et al. proposed 32-bit carry look ahead adder using approximate computing to reduce the power consumption and error metrics [14].

M. A. Akbar, B. Wang author et al. proposed self checking and repairing CLA using distributed fault detection technique and obtained parameters was compared with conventional RCA [15]. Ankit, A. V, P. Verma author et al. proposed implementation analysis of efficient adders like CSA, RAC, KSA and CLA with 8-bit, 16-bit and 32-bit designs with low power consumption, area and delay for DSP applications [5]. J. Park and Y. Kim author et al. proposed ternary carry look ahad adder using ternary value logic and implemented on FPGA [7]. S.Hibza et. al. proposed 32 bit carry look ahead adder using PTL logic and obtained parameters are compared with conventional ECARL logic [9]. C. Padma et. al. proposed efficient approximate adder by using approximate computing method for image processing applications [10].

Leakage power in digital circuits has become an important part of overall power consumption since the size of devices is decreasing [6]. In wide-bit arithmetic units, like a 32-bit Carry Look-Ahead Adder (CLA) the transistor count is very large, thus the standby leakage current is very high, even when the circuit is not operating. This paper is a review of the conventional CMOS-based CLA structures which primarily focus on speed enhancement, with little consideration of the static power dissipation which undermines the overall power efficiency in portable and high-performance systems.

To solve this problem, power gating using NMOS is taken as a viable leakage control method. Through the addition of an NMOS sleep transistor between the CLA block and ground (footer layout), the leakage pathway is broken in the standby mode, and thus the power consumption in the standby is minimized. NMOS footers are smaller in size and less resistive than PMOS header-based gating, and can be used in compact and high-speed 32-bit CLA architecture.This strategy is motivated by the need to obtain an optimum trade off between power, delay and area through the incorporation of Pass Transistor Logic (PTL) and NMOS power gating. This has reduced the number of transistors and switching capacitance and eliminated standby leakage, leading to an energy-efficient scalable 32-bit CLA design useful in modern VLSI applications.

III. CONVENTIONAL METHODOLOGY

A. PTL Logic Structure

The following structure depicted in the Figure 1 is a simple Pass Transistor Logic (PTL) switch, To pass the signal, the two transistors are activated with the control signal (e.g. $V_{CONTROL}$ =1V) and form a low-resistance circuit between V_1 and V_2. When the control signal is not active (when $V_{CONTROL}$ =0V), both transistors are turned OFF, and the input is not connected to the output [1]. This structure is a transmission gate-like structure, and is commonly employed in PTL circuits to minimize the overall number of transistors, switching capacitance, and dynamic power consumption over conventional CMOS logic.

Past CLA implementations had mostly relied on CMOS logic, which was mostly static and operated on full voltage swings [1]. Despite their dependable operation and high noise margins, the static CMOS designs typically have many transistors and therefore they have a high switching capacitance and dynamic power consumption. In order to minimize power loss, scientists considered other methods like adiabatic logic and complementary pass-transistor logic (CPL) that tries to minimize the power lost in switching transitions. Additionally, multi-threshold CMOS (MTCMOS) techniques were introduced to reduce leakage power in deep sub-micron technologies by selectively assigning high-threshold devices in non-critical paths. CLA designs focusing on sub-threshold operation were also investigated to address the growing dominance of static power in scaled technologies [2], [8].

To further reduce hardware complexity and improve speed, Pass Transistor Logic (PTL) has been adopted in arithmetic circuit design. PTL replaces conventional complementary pull-up and pull-down networks with pass transistor structures, thereby reducing transistor count and internal node capacitance. As reported in [3], PTL based arithmetic circuits achieve improved power-delay product due to reduced switching capacitance and simplified logic structures. Similarly, single-phase all-N-transistor logic approaches have been proposed to enhance speed and reduce area in CLA implementations [4].

In practical implementations, a 32-bit CLA is typically constructed using a hierarchical modular architecture. Smaller CLA blocks like 4-bit, as illustrated in Figure 2. or 8-bit modules are then designed and then cascaded to develop higher-bit adders. Group generate and group propagate signals are calculated at both levels of the hierarchy to hasten the process of evaluating carries between blocks. This modular implementation allows linear delay to be avoided with the increasing bit width and enhanced scalability when arithmetic units dealing with wide words are needed [1].

Although these are the benefits, there are some limitations to the traditional PTL based 32-bit CLA. A significant one is that there is degradation of the threshold voltages across pass transistors. Passing the logic high signal through an NMOS device, the change of the output voltage is the threshold voltage drop, which causes poor output swing and decrease in noise margins.In cascaded multi-bit structures, this degradation may accumulate and affect signal integrity. Furthermore, although PTL reduces dynamic power through lower transistor count,

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leakage power remains a significant challenge in deep sub-micron technologies [2], [8]. High-speed CMOS implementations have addressed leakage through dual-gate oxide and optimized device engineering techniques, while power gating strategies have been proposed to suppress standby leakage currents.

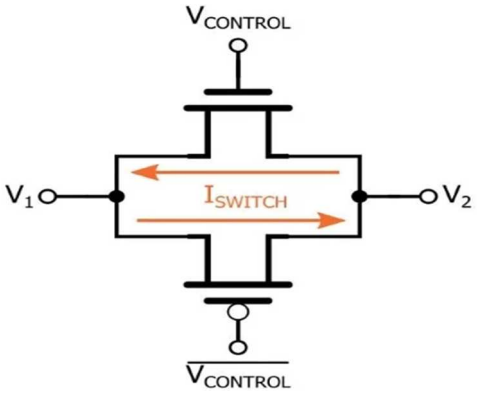


Fig. 1. PTL Structure

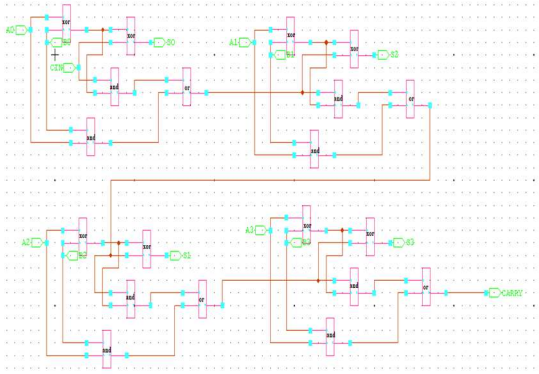


Fig. 2. PTL Based 4-Bit CLA

The Conventional PTL Based 32-bit CLA implementations as shown in Figure 3. whether based on static CMOS, adiabatic logic, multi-threshold techniques, or PTL achieve a high-speed operation and moderate hardware efficiency. However, issues like leakage power, decrease in threshold voltage, reduction in voltage swing and reliability of long path signals are also hot topics in the sophisticated technology nodes

The current 32-bit Carry Look-Ahead (CLA) adder constructed with Pass Transistor Logic (PTL) proves to be faster and fewer transistors than a traditional CMOS design, and thus is appropriate to arithmetic operations of high performance as shown in Figure 3. Reduced switching activity and area efficiency is attained by the circuit, based on the use of PTL, which helps to save moderate amounts of power and reduce carry propagation. Nevertheless, in the PTL structures because of threshold voltage drop and the constant leakage paths, the total power dissipation and voltage swing degradation may impair the performance at low-voltage regime.

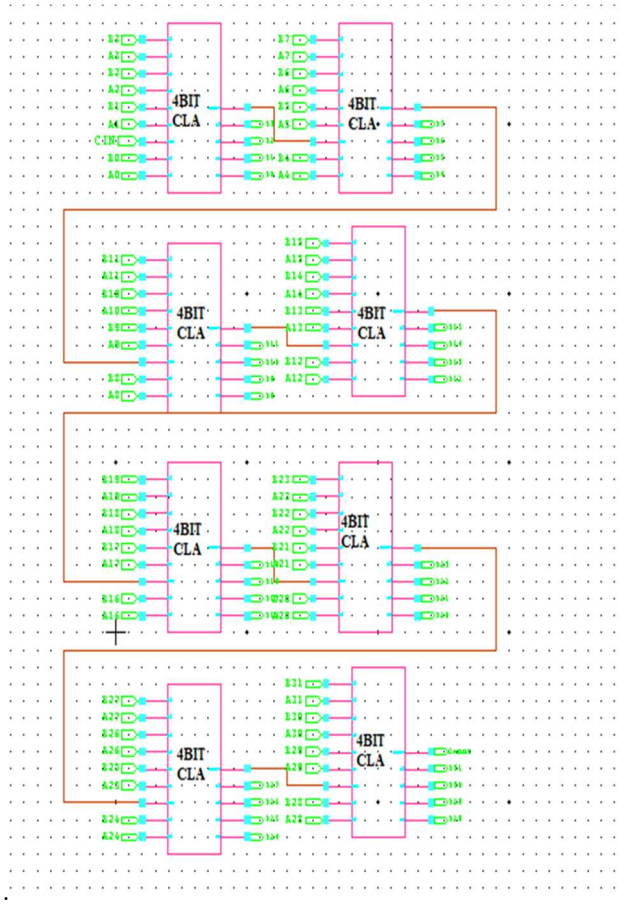


Fig. 3. Conventional PTL Based 32-Bit CLA

IV. PROPOSED METHODOLOGY

A NMOS transistor is effectively used by power gating to minimize leakage power in digital circuits when in standby is presented in Figure 4. In this approach, a NMOS transistor otherwise known as a sleep transistor is placed between the logic block and the ground terminal in a footer designation. The transistor gate of this type is driven by a sleep control signal which decides whether the circuit is in an active mode or in the standby mode. In active mode, the sleep signal activates the NMOS transistor and provides a connection with a low resistance between the logic circuit and ground. This permits the passage of normal current which allows the circuit to conduct its due operations without impairment. The logic in this state is a traditional circuit with low sensitivity to delay with a well-sized sleep transistor.

In standby, the NMOS transistor is switched off by the sleep signal and basically the logic block is disconnected to the ground. Consequently, leakage paths of the supply to ground are broken, which drastically causes the dissipation of the static power to decrease. As leakage current is noticeable in large-scale circuits that contain a large number of transistors, power gating can be used to reduce the amount of power wasted when the circuit is idle.

The NMOS footer design is desirable in most designs due to its small area and low resistance as compared to PMOS header-

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based gating. The sleep transistor should however be carefully sized to prevent high voltage drop and poor performance. In general, the idea of power gating based on NMOS offers a feasible and effective way of regulating leakage power without jeopardizing the stable work of the circuit.

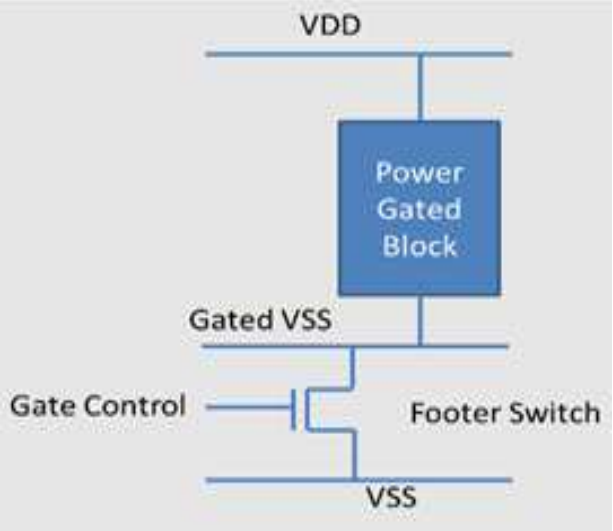


Fig. 4. Proposed NMOS Power Gated Structure

A. 32-Bit CLA using PTL and NMOS Power Gated

In the proposed 32-bit Carry Look-Ahead Adder (CLA) implemented using Pass Transistor Logic (PTL), the power-gated NMOS technique is applied to effectively reduce standby leakage power. Since a 32-bit CLA consists of a large number of transistors due to parallel carry generation and modular block implementation, leakage current becomes significant when the circuit remains idle. To address this issue, an NMOS sleep transistor is inserted as a footer device between the PTL based CLA block and ground. The sleep transistor is controlled by a sleep signal that determines the operating mode of the circuit.

During active computation, the NMOS sleep transistor is turned ON, allowing normal current flow and ensuring that the CLA performs high-speed carry generation and sum computation without functional degradation. In standby mode, the sleep transistor is turned OFF, isolating the entire 32-bit PTL based CLA from ground and thereby cutting off leakage current paths. This minimally changes the core PTL logic structure effectively reducing the overall power dissipation when at rest.

The combination of NMOS power gating and PTL schematic as shown in Figure 5, allows the design to enjoy low transistor count as well as reduced switching capacitance (via PTL) and reduced leakage power (via power gating). This is a combination that improves the overall energy saving and speed of use which does not affect high performance and the result of 32-bit CLA is better suited to low power and high performance VLSI applications.

B. NMOS Power Gating in the XOR Gate of 4-Bit CLA using PTL within the 32-Bit Architecture

The proposed design of a 32-bit CLA architecture, with NMOS power gating at the XOR gate level in each 4-bit CLA block as shown in Figure 6. XOR gate is necessary to produce the Propagate (P) signal and final Sum output, and thus it is one of the most active and power sensitive parts in the adder. It is a major source of dynamic and leakage power consumption as it involves switching and transition between signals frequently and frequently. In order to minimize leakage in idle mode, a NMOS sleep transistor has been added in the ground sub-path of the PTL XOR gate.

When the sleep transistor is in the active state, it is switched ON and gives the sleep transistor an appropriate conduction path to ground so that the XOR gate can operate normally without impacting on the carry generation or sum computation. Driving overheads and voltage drop are reduced with the correct sizing of transistors. In the standby state, the sleep transistor is switched off, and the XOR gate is not connected to ground, and leakage current in the 4-bit CLA block is minimized. The proposed architecture is designed using hierarchical modular technique, where the 32-bit CLA is comprised of cascading multiple optimized 4-bit CLA blocks as shwon in Figure 8. It is possible to provide NMOS power gating at the XOR gate level to provide distributed leakage reduction to the entire architecture. It is a focused gating methodology that reduces the static power usage and doesn't affect any of the high-speed performance or the hardware complexity required to be a reasonable fraction of the overall energy consumption of the PTL based 32-bit CLA architecture.

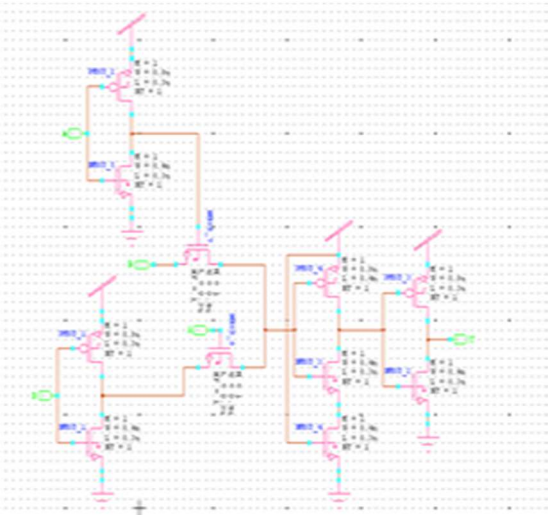


Fig. 5. Schematic of Proposed NMOS Power Gated

Similar to hierarchical arithmetic designs, each 4-bit block independently generates its propagate ($P_i=A_i \oplus B_i$) and generate ($G_i=A_i \cdot B_i$) signals and computes carry outputs using parallel look-ahead equations rather than sequential ripple propagation, thereby significantly reducing delay. The internal carry signals within each 4-bit module are derived through combi-national look-ahead expressions, and the sum outputs are obtained as $S_i=P_i \oplus C_i$.To enhance hardware efficiency, the XOR gates used for propagate and sum generation are implemented using Pass Transistor Logic (PTL), which reduces transistor count, internal node capacitance, and dynamic switching power compared to conventional CMOS logic. The simulation results of 4-Bit CLA using PTL and Power gated NMOS as shown in Figure 7. The proposed low power design has more advantages like reduced

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energy consumption, higher integration density, extended battery life and also improves the system reliability.

V. RESULTS AND DISCUSSION

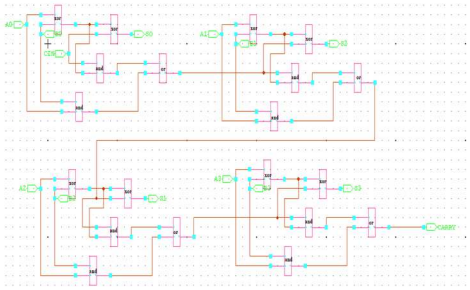


Fig. 6. Schematic of Proposed 4-Bit CLA with Power Gated NMOS

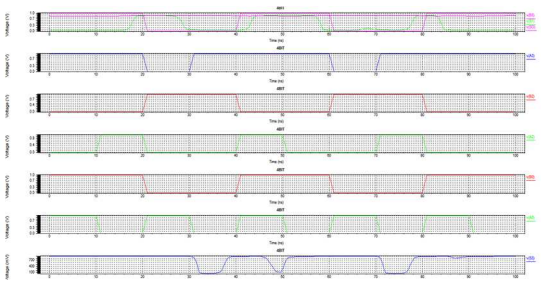


Fig. 7. Simulation Results of of Proposed 4-Bit CLA using PTL with NMOS Power Gated

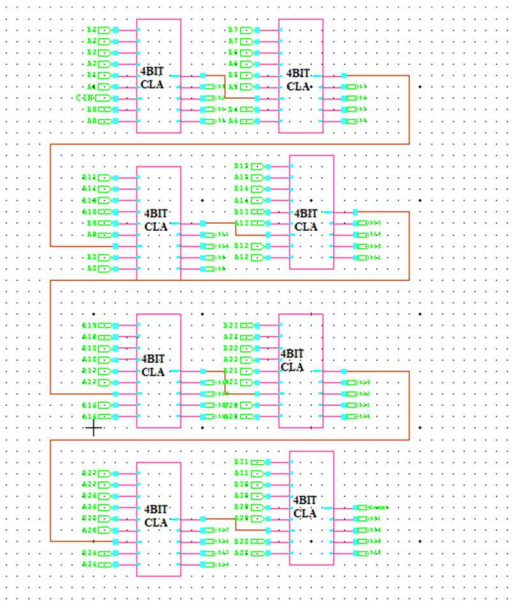


Fig. 8. Schematic of Proposed 32-Bit CLA using PTL with NMOS Power Gated

The output wave forms are the simulated sum outputs (S0 to S31) of the proposed 32-bit Carry Look-Ahead Adder which is a PTL and NMOS implementation as demonstrated in Figure 9. The switches between the logic 0 and 1 verify the proper 32-bit binary addition and effective parallel computation of the carried out computations and consequently low propagation delay. The fact that the voltage levels are clean and the glitches are minimal shows that it is operating steady and the built-in NMOS power gating minimizes leakage when in standby and does not adversely affect performance when active. In general, the findings confirm that, the proposed PTL based 32-bit CLA can be very fast and with better energy efficiency which can be used in low-power VLSI.

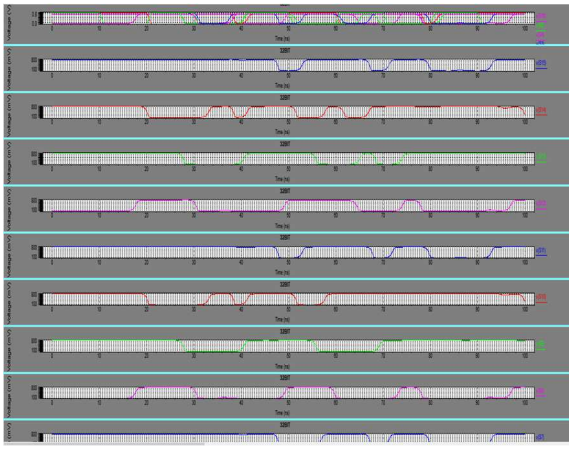


Fig. 9. Simulation Results of Proposed 32-Bit CLA using PTL with Power Gated NMOS

TABLE I. COMPARISON OF CONVENTIONAL METHODS AND PROPOSED 32-BIT CLA USING PTL WITH POWER GATED NMOS

Method	Conventional 32-Bit CLA using PTL with Transmission Gate	Proposed 32-Bit CLA using PTL with power gated NMOS
Area (No. Of Transistors)	1152	1176
Power (mW)	5.91	4.69
Delay (ns)	9.80	9.38

The performance of a Carry Look ahead Adder (CLA) written in two logic styles is compared in the Table I that create. The current design utilizes conventional Pass Transistor Logic (PTL) design, but the proposed design incorporates the PTL and power-gated NMOS based transistors. From the Table I. the proposed design can reduce the power consumption by 23.01 percent and the current power consumption will fall to 4.69mW as compared to conventional 32-Bit CLA using PTL with Transmission Gate. This has been achieved through the use of power gating technique which in essence minimizes leakage current in the idle situation. The proposed design has an improvement of 4.37 percent on its propagation delay, as the delay decreased to 9.38ns as compared to conventional 32-Bit CLA using PTL with Transmission Gate. This improvement is made by improving sizing of transistors and more efficient signal propagation paths. Nevertheless, there is a slight increment of the circuit area between 1152 and 1176 units, which constitute a 2.06% area overhead with the implementation of power-gating transistors and the control circuitry required as compared to conventional 32-Bit CLA using PTL with Transmission Gate. Although this is a slight rise, the trade-off is appeased by the high level of power saved as well as enhanced speed.

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VI. CONCLUSION

Although the power-gating circuitry added slightly to increase the number of transistors, the overhead is insignificant as compared to the large energy consumption saved. The proposed design has full voltage swing operation so that the signal integrity is not lost and can remain compatible with normal CMOS logic levels. As a result, the architecture is well balanced in terms of power, performance and complexity in design. The research out come of this paper is the proposed 32-bit CLA architecture provides a balance of improvement, with a reduction of power consumption of 23.01% and an operation delay reduction of 4.37% and a minimal increase in area of 2.06%, that it is most applicable to modern low-power VLSI designs.The proposed 32-Bit CLA adder is therefore highly suitable for low-power, high-performance VLSI applications, including portable electronics, battery-powered systems, and energy-sensitive computing platforms. The suggested future scope of the proposed design is it can be evaluated with 14nm and 7nm technologies, to further optimize the power consumption. The proposed design can be re-configurable to 64-bit and 128-bit adder for high performance and computing applications. The proposed design can also integrate as power gating with voltage scaling or power gating with clock gating methods for further optimizing the power.

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